

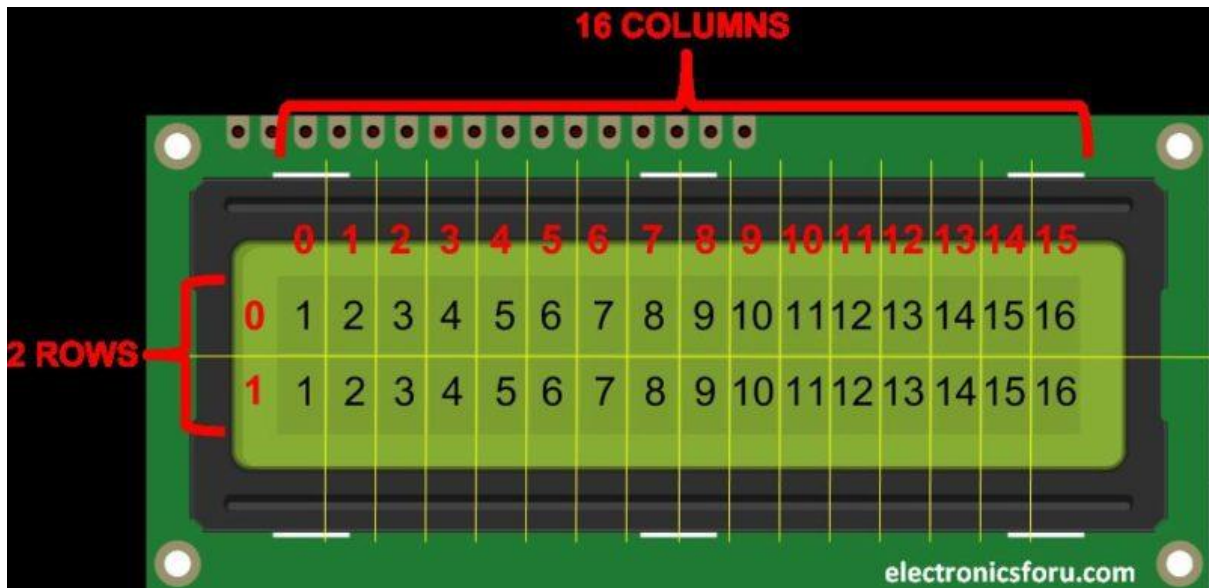
16X2 LCD Display

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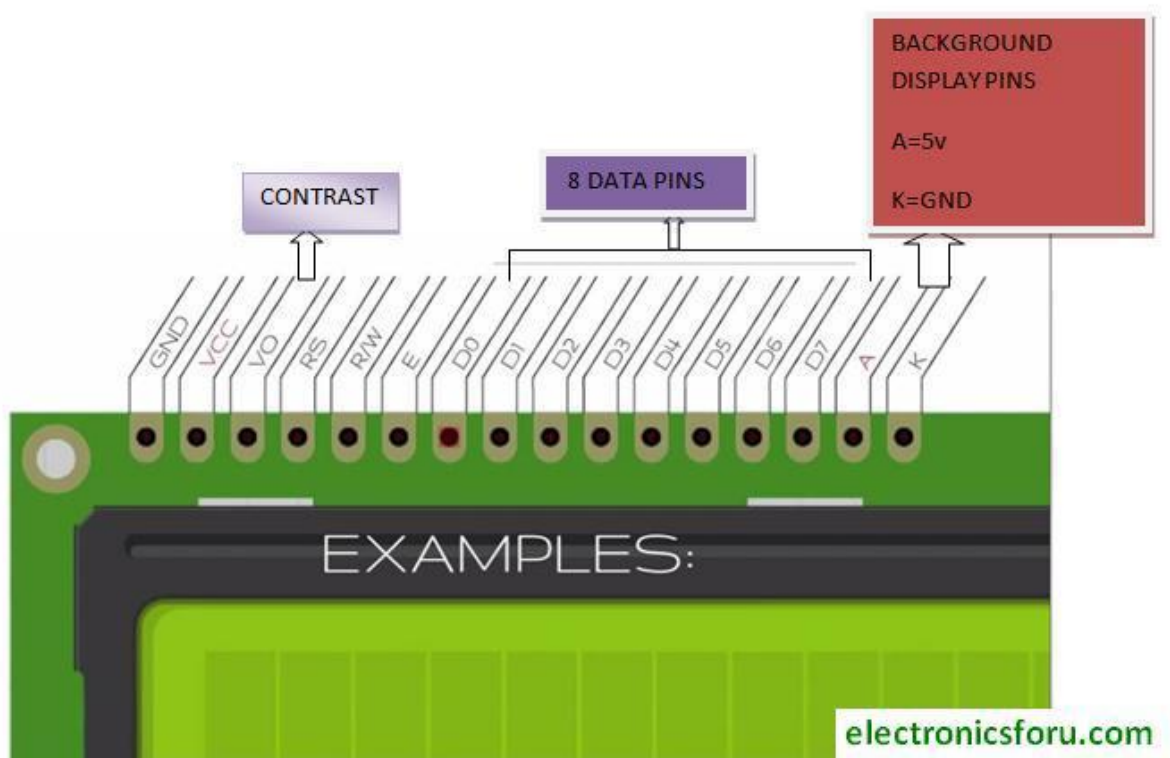
16X2 LCD Display

An LCD screen is an electronic display module that uses liquid crystal to produce a visible image. The 16×2 LCD display is a very basic module commonly used in DIYs and circuits. The 16×2 translates a display of 16 characters per line in 2 such lines. In this LCD, each character is displayed in a 5×7-pixel matrix.



LCD 16×2

16X2 LCD Display Pinout Diagram



Pin Diagram

16X2 LCD Display

Pin No.	Function	Name - Advertisement -
1	Ground (0V)	Ground
2	Supply voltage; 5V (4.7V – 5.3V)	VCC
3	Contrast adjustment; the best way is to use a variable resistor such as a potentiometer. The output of the potentiometer is connected to this pin. Rotate the potentiometer knob forward and backward to adjust the LCD contrast.	Vo / VEE
4	Selects command register when low, and data register when high	RS (Register Select)
5	Low to write to the register; High to read from the register	Read/write
6	Sends data to data pins when a high to low pulse is given; Extra voltage push is required to execute the instruction and EN (enable) signal is used for this purpose. Usually, we set en =0, when we want to execute the instruction, we make it high en=1 for some milliseconds. After this we again make it ground that is, en=0.	Enable
7	8-bit data pins	DB0
8		DB1
9		DB2
10		DB3
11		DB4
12		DB5
13		DB6
14		DB7
15	LED Backlight VCC (5V)	Led+
16	LED Backlight Ground (0V)	Led-

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RS (Register Select):

A 16X2 LCD has two registers, namely, command and data. The register select is used to switch from one register to other. RS=0 for the command register, whereas RS=1 for the data register.

Command Register:

The command register stores the command instructions given to the LCD. A command is an instruction given to an LCD to do a predefined task. Examples like:

- initializing it
- clearing its screen
- setting the cursor position
- controlling display etc.

Processing for commands happens in the command register.

Data Register:

The data register stores the data to be displayed on the LCD. The data is the ASCII value of the character to be displayed on the LCD. When we send data to LCD, it goes to the data register and is processed there. When RS=1, the data register is selected.

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Important Command Codes for 16×2 LCD

Sr.No.	Hex Code	Command to LCD instruction Register
1	01	Clear display screen
2	02	Return home
3	04	Decrement cursor (shift cursor to left)
4	06	Increment cursor (shift cursor to right)
5	05	Shift display right
6	07	Shift display left
7	08	Display off, cursor off
8	0A	Display off, cursor on
9	0C	Display on, cursor off
10	0E	Display on, cursor blinking
11	0F	Display on, cursor blinking
12	10	Shift cursor position to left
13	14	Shift the cursor position to the right
14	18	Shift the entire display to the left
15	1C	Shift the entire display to the right
16	80	Force cursor to the beginning (1st line)
17	C0	Force cursor to the beginning (2nd line)
18	38	2 lines and 5×7 matrix

In the table above you can see starting addresses for each character with their printing commands.

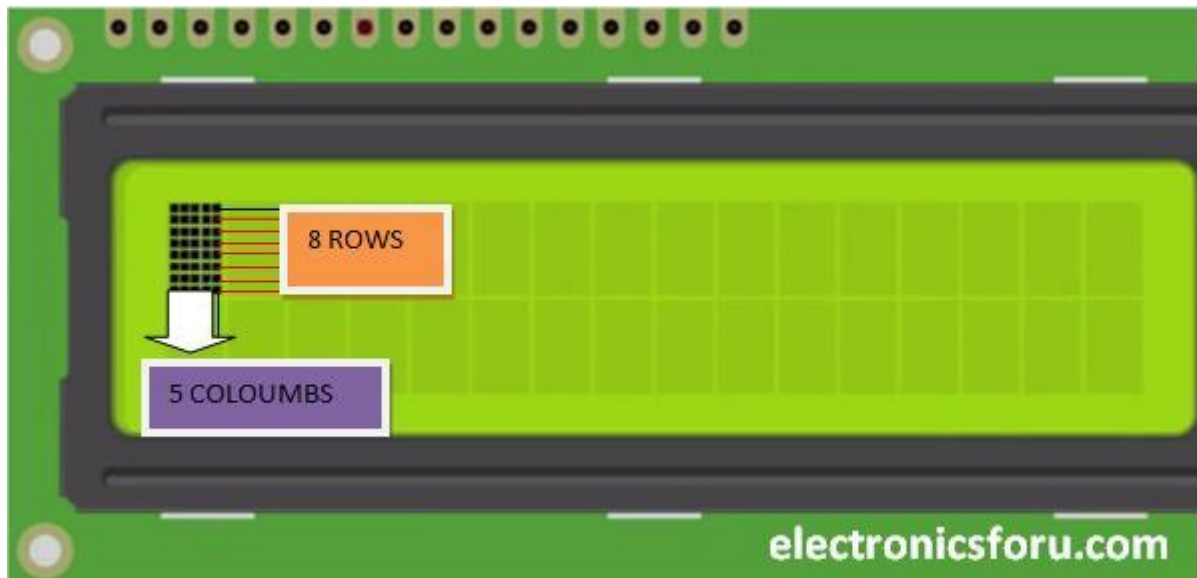
The first character is generated at addresses 0x40 to 0x47 and is printed on LCD by just sending a command 0.

The second character is generated at addresses 0x48 to 0x55 and is printed by sending a command 1.

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How to Generate Custom Characters in CG-RAM

In LCD displays, each character is in a 5×8 matrix. Where 5 is the number of columns and 8 is the number of rows.



LCD 16×2 Matrix

Here is a simple example of how to create the letter 'b' in CG-RAM.

The Array for generating 'b' is `char b [7] = {0x10,0x10,0x16,0x19,0x11,0x11,0x1E};` That is,

- Send the address where you want to create a character.
- Now create your character at this address. Send the 'b' character array values defined above one by one to the data register of the LCD.
- To print the generated character at 0x40. Send command 0 to the command register of LCD. The table below would explain this more clearly-

CG-RAM ADDRESS	Byte Address	C1	C2	C3	C4	C5	Pattern for b is	
Starting Address of CG-RAM.	0 0 0 0	1	0	0	0	0	Row1	0x10=00010000
	0 0 0 1	1	0	0	0	0	Row2	0x10=00010000
	0 0 1 0	1	0	1	1	0	Row3	0x16=00010110
	0 0 1 1	1	1	0	0	1	Row4	0x19=00011001
	0 1 0 0	1	0	0	0	1	Row5	0x11=00010001
	0 1 0 1	1	0	0	0	1	Row6	0x11=00010001
	0 1 1 0	1	1	1	1	0	Row7	0x1E=00011110
	0 1 1 1	0	0	0	0	0		

Hexadecimal=0x40
Decimal=64

First three bits are treated as Dont cares.
* Rest five bits are loaded at the address

Cursor position

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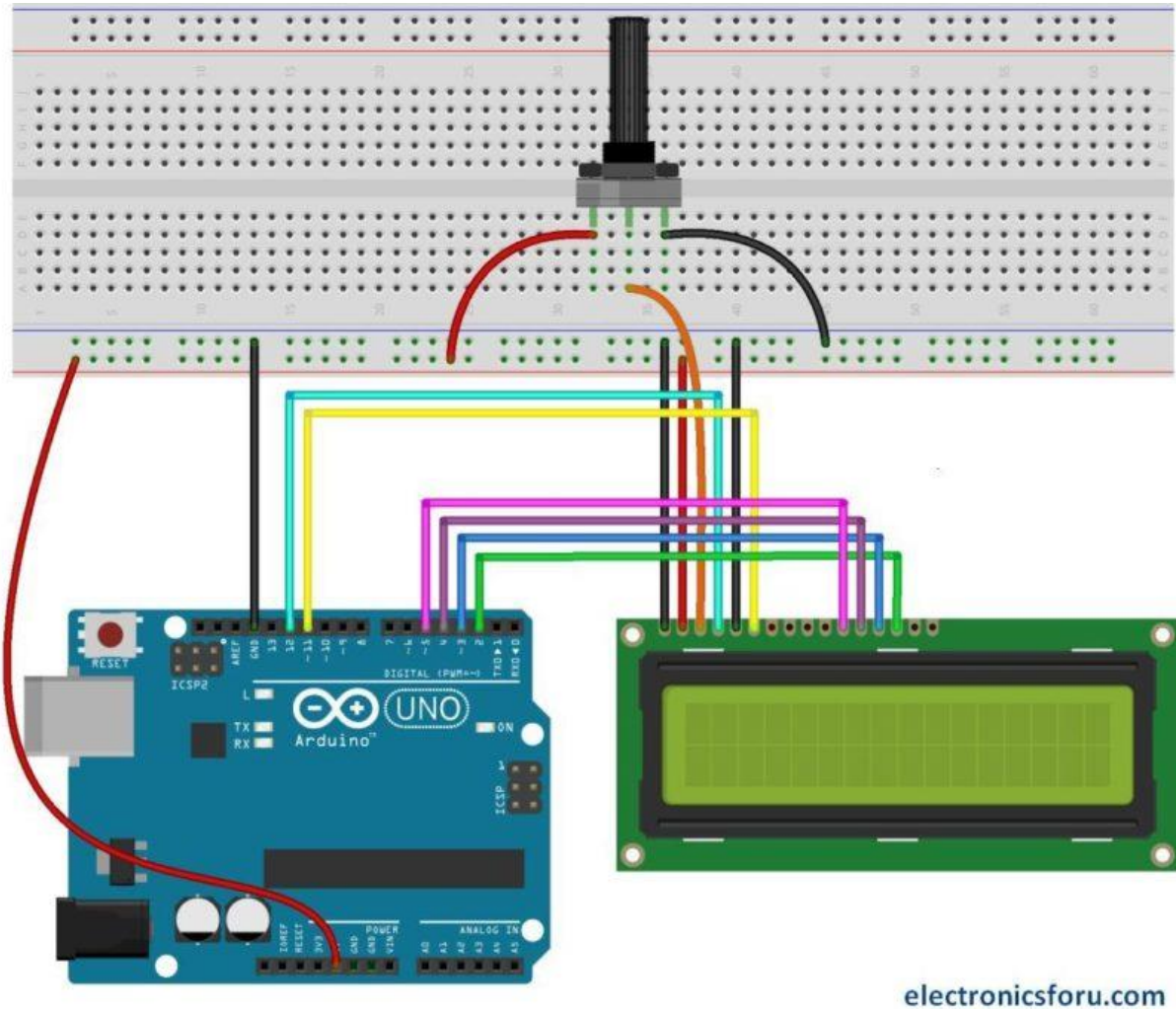
CG-RAM Address Table

Interfacing LCD 16X2 with Arduino

LCD modules are very important in many Arduino-based embedded system designs to improve the user interface of the system. Interfacing with Arduino gives the programmer more freedom to customize the code easily.

16X2 LCD Display

Any cost-effective Arduino board, a 16X2 character LCD display, jumper wires, and a breadboard are sufficient enough to build the circuit.



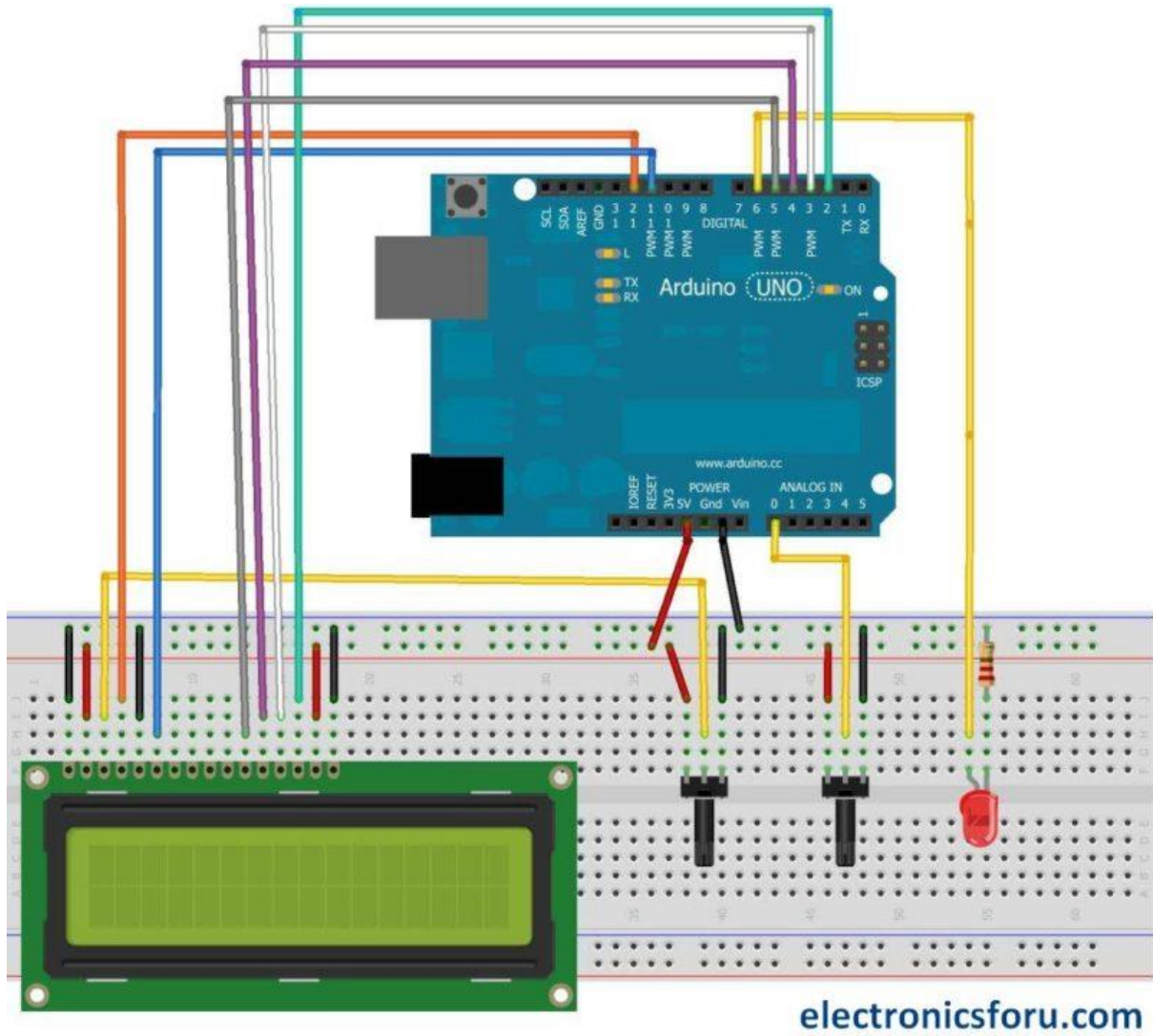
Interfacing Circuit

For detailed explanation, you can check how to use LCD with Arduino.

Display the LED Brightness on a 16×2 LCD

The combination of an LCD and Arduino yields several projects, the simplest one being LCD to display the LED brightness. All we need for this circuit is an LCD, Arduino, breadboard, a resistor, potentiometer, LED, and some jumper cables. The circuit connections are below.

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LCD 16×2 Project Circuit